

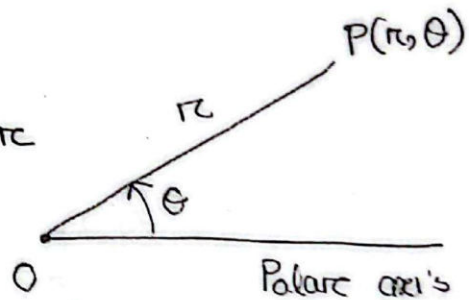
lec-11

## Polar Coordinate System :

H-A. sec-10.2

A polar coordinate system in a plane consists of a fixed point  $O$ , called the pole (or origin) and a ray emanating from the pole, called the polar axis. In such a coordinate system we can associate with each point  $P$  in the plane a pair of polar coordinates  $(r, \theta)$  where  $r$  is the distance from  $P$  to the pole and  $\theta$  is an angle from the polar axis

to the ray  $OP$ . The number  $r$  is called the radial coordinate of  $P$  and the number



$\theta$  the angular coordinate (or polar angle) of  $P$ .

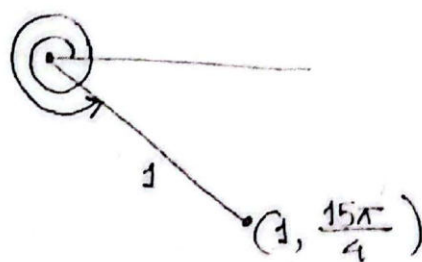
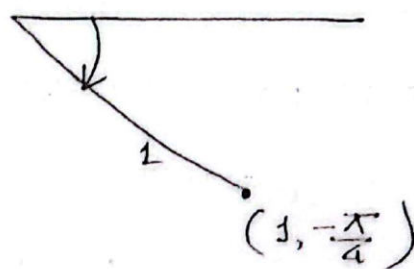
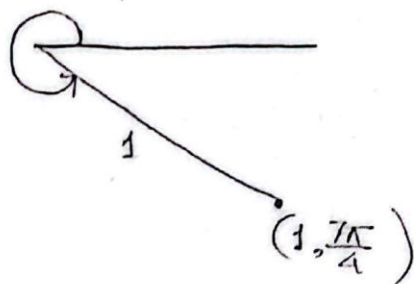
direction

\*  $\theta$  is positive in the counterclockwise direction from the polar axis.

\*  $\theta$  is negative in the clockwise direction from the polar axis.

The polar coordinates of a point are not unique. For example, the polar coordinates

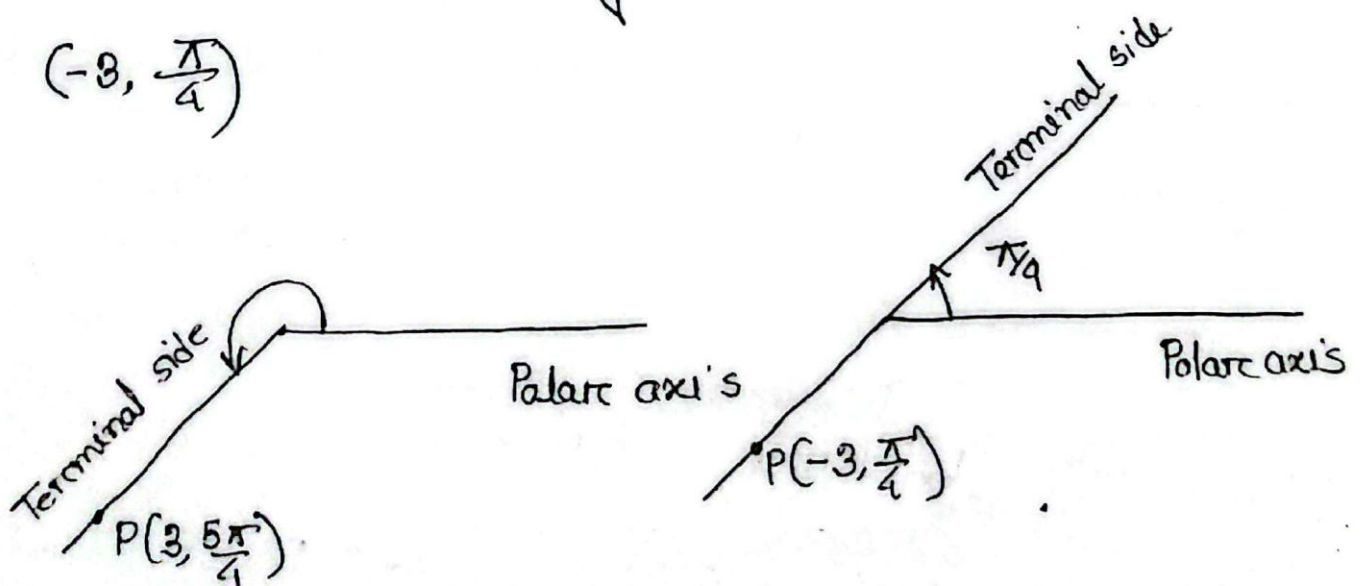
$(1, \frac{7\pi}{4})$ ,  $(1, -\frac{\pi}{4})$  and  $(1, \frac{15\pi}{4})$  all represent the same point.



In general, if point  $P$  has polar coordinates  $(r, \theta)$  then  $(r, \theta + 2n\pi)$  and  $(r, \theta - 2n\pi)$  are also polar coordinates of  $P$  for any nonnegative integer  $n$ . Thus every point has infinitely many pairs of polar coordinates.



The radial coordinate  $r$  of a point  $P$  is nonnegative, since it represents the distance from  $P$  to the pole. However, it will be convenient to allow nonnegative values for  $r$  as well. To motivate an appropriate definition we can reach the point  $P(3, \frac{5\pi}{4})$  by rotating the polar axis through an angle of  $\frac{5\pi}{4}$  and then moving 3 units from the pole along the terminal side of the angle or we can reach the point  $P$  by rotating the polar axis through an angle of  $\frac{\pi}{4}$  and then moving 3 units from the pole ~~on~~ along the extension of the terminal side. This suggests that the point  $(3, \frac{5\pi}{4})$  might also be denoted by  $(-3, \frac{\pi}{4})$



In general, the terminal side of the angle  $\theta + \pi$  is the extension of the terminal side of  $\theta$ , therefore  $(-\pi, \theta)$  and  $(\pi, \theta + \pi)$  are polar coordinates of the same point.

### Relationship Between Polar and Rectangular Coordinates:

Polar  $\rightarrow$  Rectangular  
 $(r, \theta)$   $(x, y)$

Rectangular  $\rightarrow$  Polar  
 $(x, y)$   $(r, \theta)$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

Example: Find the rectangular coordinates of the point P whose polar coordinates are  $(6, \frac{2\pi}{3})$ .

Solution:

We know that

$$x = r \cos \theta = 6 \cos\left(\frac{2\pi}{3}\right) = -3$$

$$y = r \sin \theta = 6 \sin\left(\frac{2\pi}{3}\right) = 3\sqrt{3}$$

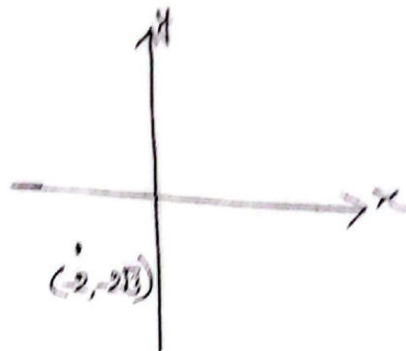


Therefore, the rectangular coordinates of P are  $(-3, 3\sqrt{3})$ .

Example: Find polar coordinates of the point P whose rectangular coordinates are  $(-2, -2\sqrt{3})$ .

Solution: We know that

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-2)^2 + (-2\sqrt{3})^2} \\ &= \sqrt{4 + 12} = 4 \end{aligned}$$



$$\tan \theta = \frac{y}{x}$$

$$\Rightarrow \tan \theta = \frac{-2\sqrt{3}}{-2}$$

$$\Rightarrow \tan \theta = \sqrt{3}$$

The point  $(-2, -2\sqrt{3})$  lies in the third quadrant, it follows that the angle satisfying the requirement  $0 \leq \theta < 2\pi$  is  $\theta = \frac{4\pi}{3}$ . Thus  $(r, \theta) = (4, \frac{4\pi}{3})$

## Cylindrical Coordinate System :-

11.8 (H-Anton)

Three coordinates are required to establish the location of a point in 3-space. In a rectangular coordinate system, the coordinates can be any real numbers, but in cylindrical and spherical coordinate systems there are restrictions on the allowable values of the coordinates.

### Relationship between Cartesian Rectangular and Cylindrical Coordinates :-

Cylindrical  $\rightarrow$  Cartesian Rectangular

$$(r, \theta, z) \rightarrow (x, y, z)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Cartesian Rectangular  $\rightarrow$  Cylindrical

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

$$z = z$$

Example:- Find the cylindrical coordinates of point P whose rectangular coordinates are Cartesian  $(4, -4, 4\sqrt{6})$ .

Solution: We know that

$$\begin{aligned} \text{ii. } \sqrt{x^2+y^2} &= \sqrt{(4)^2+(-4)^2} = \sqrt{16+16} \\ &= \sqrt{32} = 4\sqrt{2} \end{aligned}$$

$$\tan \theta = \frac{y}{x} \Rightarrow \tan \theta = \frac{-4}{4} \Rightarrow \tan \theta = -1$$

$$\Rightarrow \theta = \frac{7\pi}{4}$$

[since the given point has a negative y-coordinate]

$$z = 4\sqrt{6}$$

$\therefore$  The cylindrical coordinates are  $(4\sqrt{2}, \frac{7\pi}{4}, 4\sqrt{6})$ .

Example: Find the cylindrical coordinates of the point P whose <sup>cartesian</sup> ~~rectangular~~ coordinates are  $(4\sqrt{2}, \frac{7\pi}{4}, 4\sqrt{6})$ .

Solution: We know that,

$$x = r \cos \theta$$

$$= 4\sqrt{2} \cos\left(\frac{7\pi}{4}\right) = 4\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 4$$



$$\begin{aligned}
 y &= r \sin \theta \\
 &= 4\sqrt{2} \sin\left(\frac{7\pi}{4}\right) \\
 &= 4\sqrt{2} \cdot \left(\frac{-1}{\sqrt{2}}\right) \\
 &= -4
 \end{aligned}$$

$$z = z = 4\sqrt{6}$$

Therefore, the cylindrical coordinates are  $(4, -4, 4\sqrt{6})$ .

Transformation of Equation :-

Cartesian  
~~Rectangular~~  $\rightarrow$  cylindrical

$$2x + 3y + 4z = 1$$

$$\Rightarrow 2r \cos \theta + 3r \sin \theta + 4z = 1$$

$$\Rightarrow 4z = 1 - 2r \cos \theta - 3r \sin \theta$$

$$\Rightarrow z = \frac{1 - 2r \cos \theta - 3r \sin \theta}{4}$$

✱



cylindrical  $\rightarrow$  Rectangular

Then

$$r^2 \cos 2\theta = z$$

$$r^2 \cos 2\theta = z$$

$$\Rightarrow r^2 (1 + 2 \cos^2 \theta - 1) = z$$

$$\Rightarrow r^2 (2 \cos^2 \theta - 1) = z$$

$$\Rightarrow r^2 + 2 r^2 \cos^2 \theta = z$$

$$\Rightarrow 2 r^2 \cos^2 \theta - r^2 = z$$

$$\Rightarrow r^2 + 2 (r \cos \theta)^2 = z$$

$$\Rightarrow 2x^2 - (x^2 + y^2) = z$$

$$\Rightarrow x^2 + y^2 + 2x^2 = z$$

$$\Rightarrow x^2 - y^2 = z$$

Spherical Coordinates :- 11.8 (H. Anton)

Let's recall the cylindrical coordinates.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Now we substitute above  $r = \rho \sin \phi$  and  $z = \rho \cos \phi$  in these equation.

Then we get,

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Cartesian  
Rectangular  $\rightarrow$  Spherical  
(x, y, z) (ρ, φ, θ)

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\tan \theta = \frac{y}{x}$$

Spherical  $\rightarrow$  Cartesian  
Rectangular

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Example:- Find the <sup>Cartesian</sup> ~~rectangular~~ coordinates of the point with spherical coordinates

$$(4, \frac{\pi}{4}, \frac{\pi}{3})$$

Solution:- We know that

$$x = \rho \sin \phi \cos \theta$$

$$= 4 \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) = \sqrt{2}$$

$$y = \rho \sin \phi \sin \theta = 4 \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{3}\right) = \sqrt{6}$$

$$z = \rho \cos \phi = 4 \cos\left(\frac{\pi}{4}\right) = 2\sqrt{2}$$

The ~~rectangular~~ <sup>cartesian</sup> coordinates of the point are  $(\sqrt{2}, \sqrt{6}, 2\sqrt{2})$ .

Example :- Find the spherical coordinates of the point whose ~~rectangular~~ <sup>cartesian</sup> coordinates are  $(\sqrt{2}, \sqrt{6}, 2\sqrt{2})$ .

Solution :- We know that,

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{2 + 6 + 8} = 4$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = \frac{2\sqrt{2}}{4}$$

$$\Rightarrow \cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = \frac{\pi}{4}$$

$$\tan \theta = \frac{y}{x} \Rightarrow \tan \theta = \frac{\sqrt{6}}{\sqrt{2}}$$

$$\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$$

$\therefore$  The spherical coordinates are  $(4, \frac{\pi}{4}, \frac{\pi}{3})$ .



## Transformation of Equations :-

Conversion

~~Rectangular~~  $\rightarrow$  Spherical

$$x^2 + y^2 - z^2 = 1$$

$$\Rightarrow x^2 + y^2 + z^2 - z^2 = 2z^2 + 1$$

$$\Rightarrow \rho^2 x^2 + y^2 + z^2 = 2z^2 + 1$$

$$\Rightarrow \rho^2 = 2z^2 + 1$$

$$\Rightarrow \rho^2 = 2\rho^2 \cos^2 \phi + 1$$

$$\Rightarrow 1 = 2 \cos^2 \phi$$

$$x^2 + y^2 - z^2 = 1$$

$$\Rightarrow \rho^2 \sin^2 \phi \cos^2 \theta$$

$$+ \rho^2 \sin^2 \phi \sin^2 \theta - \rho^2 \cos^2 \phi = 1$$

$$\Rightarrow \rho^2 \sin^2 \phi - \rho^2 \cos^2 \phi = 1$$

$$\Rightarrow \rho^2 \cos^2 \phi = -1$$

~~Spherical~~  $\rightarrow$

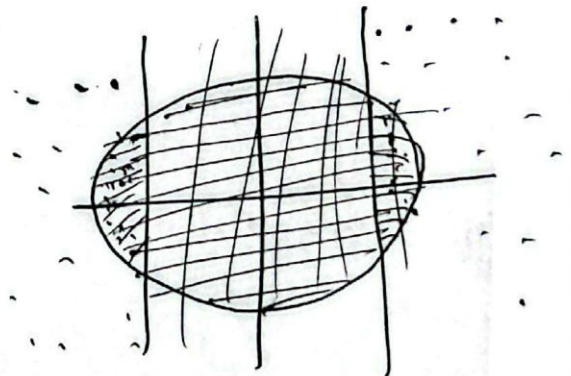
Spherical  $\rightarrow$  ~~Rectangular~~ Cartesian

$$\rho - 2 \sin \phi \cos \theta = 0$$

$$\Rightarrow \rho^2 - 2\rho \sin \phi \cos \theta = 0$$

$$\Rightarrow \rho^2 - 2x = 0$$

$$\Rightarrow x^2 + y^2 + z^2 - 2x = 0$$



### Practice Problem $\rightarrow$

11-8  $\rightarrow$  1, 2, 3, 4, 5, 6, 7, 8, 19-26,  
27-34, 35-46

Spherical to cylindrical

$$r = \rho \sin \phi, \quad \theta = \theta, \quad z = \rho \cos \phi$$

Cylindrical to spherical

$$\rho = \sqrt{r^2 + z^2}, \quad \theta = \theta, \quad \tan \phi = \frac{r}{z}$$

Example: Find the spherical coordinates of the point whose cylindrical coordinates are  $(\sqrt{3}, \frac{\pi}{6}, 3)$

Solution: We know that,

$$\rho = \sqrt{r^2 + z^2}$$

$$= \sqrt{(\sqrt{3})^2 + (3)^2} = \sqrt{12}$$

$$\theta = \frac{\pi}{6}$$

$$\tan \phi = \frac{r}{z} \Rightarrow \phi = \tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$$

The spherical coordinates of the point is,  $(\sqrt{12}, \frac{\pi}{6}, \frac{\pi}{4})$

Example: Find the cylindrical coordinates of the point whose spherical coordinates are  $(5, \frac{\pi}{4}, \frac{2\pi}{3})$ .